Ultrasound in Medicine & Biology 000 (2026) 1−8

Contents lists available at ScienceDirect

Ultrasound in Medicine & Biology

journal homepage: www.elsevier.com/locate/ultrasmedbio

Original Contribution

Automatic Skull-Template Alignment Without a Guidance Image

Oscar Bates [a,b,*], Carlos Cueto [a,b], Ciaran Coleman [b], Cameron A.B. Smith [c], Lluis Guasch [a,b], Oscar Calderon Agudo [a,b]

[a] *Department of Earth Science and Engineering, Imperial College London, London, UK*
[b] *Sonalis Imaging Limited, London, UK*
[c] *Department of Bioengineering, Imperial College London, London, UK*

A R T I C L E I N F O

Keywords:
Transcranial ultrasound
Ultrasound tomography
Full-waveform inversion (FWI)
Guidance-free registration
Neuroimaging

A B S T R A C T

Objective: Transcranial ultrasound faces significant challenges due to the human skull, which limits both imaging and therapeutic applications. High-fidelity numerical simulations can mitigate skull-induced distortions but require precise skull templates (typically derived from computed tomography scans) and accurate spatial alignment with the patient's anatomy. Current alignment methods rely on concurrent magnetic resonance imaging (MRI) for registration, introducing financial, logistical and throughput barriers.
Methods: We present manifold optimisation for full-waveform inversion (MOFI), a novel method to register skull templates without requiring a guidance image. MOFI aligns the skull template by minimising the difference between the simulated and the observed radiofrequency acoustic data.
Results: We demonstrate that MOFI accurately recovers the position of skull templates in both *in silico* and *in vitro* settings, offering a viable alternative to MRI-guided registration.
Conclusions: These results suggest that MOFI has the potential to serve as a practical alternative to MRI-guided approaches, thereby reducing barriers to the wider clinical adoption of transcranial ultrasound examination.

Introduction

Transcranial ultrasound imaging encompasses numerous therapeutic and imaging applications. Currently, the major neuroimaging modalities —magnetic resonance imaging (MRI) and X-ray computed tomography (CT) scans—are large, expensive, immobile, high-power instruments that are deployable only in specialised facilities. Pulse-echo ultrasound imaging is safe, fast and portable, but transcranial B-mode imaging struggles with limited resolution [1]. Therapeutic ultrasound imaging has few direct analogues in clinical use and encompasses neuromodulation [2], drug delivery [3] and ablation [4].

Ultrasound neurotechnologies are hindered by the high impedance and heterogeneous structure of the skull, which aberrates, scatters and attenuates the ultrasound wavefield [5]. These effects substantially degrade both therapeutic targeting [6] and transcranial imaging [7].

Recent work combining wide-aperture transcranial ultrasound tomography systems with full-waveform inversion (FWI) has shown promise for recovering intracranial structure [7−9]. FWI formulates image reconstruction as a non-linear optimisation problem, estimating the acoustic parameter field that best matches measured radiofrequency acoustic data. However, the large impedance contrast between bone and soft tissue continues to pose severe challenges for conventional gradient-based optimisation. FWI is high dimensional, often involving tens to hundreds of millions of elements, and is prone to convergence issues such as cycle skipping that impair the reconstruction [10].

Various strategies have been proposed to address these limitations, including alternative loss functions [11,12] and preconditioning the image vector via neural networks [13] or finite element grids [14], although methods these have yet to be validated on experimental ultrasound tomography data. In therapeutic ultrasound imaging, a longstanding solution is to estimate the skull's acoustic properties from patient-specific CT data and numerically model skull-induced aberrations [15], enabling corrections to be applied. One analogous approach in transcranial ultrasound tomography is to use prior information from a CT scan or MRI to derive a skull template and simplify the optimisation problem [7,8].

However, an important practical challenge is the alignment of skull templates to patient anatomy. Existing approaches typically rely on MRI guidance images to register CT-derived skull models [16], which undermines the clinical utility of transcranial ultrasound tomography and complicates deployment. Guidance-free registration remains largely unexplored, with only one prior study using reflected ultrasound time-of-flight data and constrained optimisation, restricting rotations and translations to $\pm 3^{\circ}$ and ± 3 mm [17].

We present manifold optimisation for FWI (MOFI), a framework for guidance-free skull template alignment utilising both reflected and

* Corresponding author. Department of Earth Science and Engineering, Imperial College London, London, SW7 2AZ, UK.
E-mail address: o.bates18@imperial.ac.uk (O. Bates).

https://doi.org/10.1016/j.ultrasmedbio.2026.05.003
Received 11 February 2026; Revised 16 April 2026; Accepted 5 May 2026

transmitted acoustic data, using unconstrained optimisation. MOFI identifies a smooth, low-dimensional manifold that embeds key properties of the FWI image vector, enabling optimisation over physically meaningful transformations while preserving high-fidelity wave propagation modelling. By operating in this reduced space, MOFI effectively preconditions the FWI gradient and improves robustness to large initial misalignments.

Methods

In standard FWI, the gradient is computed for each pixel of the discretised sound speed grid using an adjoint state method. The geometry of this gradient is inherently tied to the discretisation of the sound speed grid, where the dimensionality of the update is determined by the number of pixels in the image. However, this geometry is not necessarily optimal for FWI updates, because the sound speed discretisation is primarily chosen to ensure the stability of numerical simulations rather than to reflect the geometric properties of the underlying object. We hypothesised that a more geometrically informed choice of update parameters can significantly improve the update direction, enabling access to solutions that are unattainable through standard FWI. This principle underpins MOFI, which leverages a low-dimensional, geometrically meaningful parameterisation to enhance the efficiency and effectiveness of the inversion process.

FWI

We consider a standard time-domain acoustic FWI setup. The unknown is the sound−speed distribution, $\mathbf{c}(\mathbf{x})$, $\mathbf{x}\in\Omega\subset\mathbb{R}^n$. Given appropriate initial and boundary conditions, including a source $s(\mathbf{x},t)$, the acoustic wavefield $\mathbf{u}_p(\mathbf{x},t)$ satisfies

$$\Box(\mathbf{c})\mathbf{u}_p(\mathbf{x},t)=s(\mathbf{x},t),$$

where

$$\Box(\mathbf{c})=\frac{1}{\mathbf{c}(\mathbf{x})^2}\frac{\partial^2}{\partial t^2}-\nabla^2.$$

For a point source at position $\mathbf{x}_s$ with time signal $\varsigma(t)$, we write

$$s(\mathbf{x},t)=\delta(\mathbf{x}-\mathbf{x}_s)\ \zeta(t).$$

The predicted data are obtained by sampling the wavefield at receiver positions $\mathbf{x}_r$:

$$\mathbf{d}_p(\mathbf{x}_r,t;\mathbf{c},\mathbf{x}_s,\varsigma)=\delta(\mathbf{x}-\mathbf{x}_r)\mathbf{u}_p(\mathbf{x},t;\mathbf{c},\mathbf{x}_s,\varsigma).$$

Let $L(\mathbf{c})$ denote the collection of all predicted data for a given model $\mathbf{c}$ (stacking all receivers, times and sources into a single data vector), and let $\mathbf{d}$ be the corresponding observed data. We use the usual least-squares data misfit

$$f(\mathbf{c})=\frac{1}{2}\|\mathbf{d}-L(\mathbf{c})\|_2^2.$$

The goal of FWI is to find a model c that minimises $f(\mathbf{c})$. This is typically done by iterative gradient-based optimisation. The gradient of f with respect to $\mathbf{c}$ is obtained using the adjoint-state method. The gradient is given by:

$$\frac{\partial f}{\partial \mathbf{c}}=-\mathbf{u}_p(\mathbf{c})^T\frac{\partial\Box(\mathbf{c})}{\partial \mathbf{c}}\mathbf{u}_{\mathrm{adj}}(\mathbf{c}),$$

where the adjoint wavefield $\mathbf{u}_{\mathrm{adj}}(\mathbf{c})$ is the solution of the adjoint problem:

$$\Box(\mathbf{c})^T\mathbf{u}_{\mathrm{adj}}(\mathbf{c})=L(\mathbf{c})-\mathbf{d}.$$

In practice, each iteration consists of:

1. Forward solve: Compute $\mathbf{u}_p$ and the predicted data $L(\mathbf{c})$.
2. Adjoint solve: Back-propagate the data residual $L(\mathbf{c})-\mathbf{d}$ to obtain an adjoint wavefield $\mathbf{u}_{\mathrm{adj}}$.
3. Gradient computation: Form the zero-lag of the cross-correlation between the forward and adjoint wavefields to obtain the gradient vector $\partial f/\partial\mathbf{c}$ at each grid point.

The FWI gradient indicates how the sound−speed model should be updated to reduce the misfit. In conventional FWI, we use this gradient directly to perform a pixel-wise update of $\mathbf{c}$.

Manifold optimisation for FWI

The standard pixel-wise parameterisation in FWI is inefficient when the mismatch between predicted and observed data arises from geometric deformations, such as global translations or rotations. To address this issue, MOFI introduces a low-dimensional parameterisation that captures the geometry of the deformation. In this work, we focus on representing the model update through rigid body motion—namely, translations and rotations of the image—although MOFI can also be extended to nonrigid deformations and other parametrisations. By constraining updates to a geometrically meaningful manifold, MOFI improves the efficiency, accuracy and robustness of the inversion process.

We start from a reference sound−speed image $\mathbf{c}(\mathbf{x}_{\mathrm{orig}})$ defined on the regular grid $\mathbf{x}_{\mathrm{orig}}$. Rather than updating $\mathbf{c}$ directly, we introduce a small set of motion parameters:

$$\boldsymbol{\phi}=\{\theta,\delta_1,\delta_2\},$$

which describe a rotation by angle θ around a chosen centre and a translation by distance (δ_1,δ_2).

These parameters define a rigid transformation of coordinates using the special Euclidean matrix group $SE(2)$:

$$\begin{aligned}\mathbf{x}'_{\mathrm{orig}} &= T_{\boldsymbol{\phi}}\left(\mathbf{x}_{\mathrm{orig}}\right)\\ \begin{pmatrix}x'_{\mathrm{orig},1}\\ x'_{\mathrm{orig},2}\\ 1\end{pmatrix} &= \begin{pmatrix}\cos(\theta) & -\sin(\theta) & \delta_1\\ \sin(\theta) & \cos(\theta) & \delta_2\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}x_{\mathrm{orig},1}\\ x_{\mathrm{orig},2}\\ 1\end{pmatrix}.\end{aligned}$$

Here, T_ϕ is the forward transformation matrix that maps the original coordinates $\mathbf{x}_{\mathrm{orig}}$ to the transformed coordinates $\mathbf{x}'_{\mathrm{orig}}$.

The aim is to find pixel values for the transformed image. The transformed coordinates $\mathbf{x}'_{\mathrm{orig}}$ are not usually integers, so $\mathbf{c}_\phi(\mathbf{x}'_{\mathrm{orig}})$ is not defined on a regular pixelated grid. Instead, a pixelated coordinate grid $\mathbf{x}_{\mathrm{trans}}$ is defined for a new image $\mathbf{c}(\mathbf{x}_{\mathrm{trans}})$. Then, each pixel of the new image is associated with a location $\mathbf{x}'_{\mathrm{trans}}$ on the original image

$$\mathbf{x}'_{\mathrm{trans}}=T_\phi^{-1}(\mathbf{x}_{\mathrm{trans}}),$$

where the inverse transformation T_ϕ^{-1} maps the new coordinate system back to the original coordinate system.

The transformed sound−speed image is then parameterised in terms of the original image, given by:

$$\mathbf{c}_\phi(\mathbf{x}'_{\mathrm{trans}})=\mathbf{c}\left(T_\phi^{-1}(\mathbf{x}_{\mathrm{trans}})\right).$$

Intuitively, $\mathbf{c}_\phi(\mathbf{x}'_{\mathrm{trans}})$ represents the original sound−speed image $\mathbf{c}(\mathbf{x}_{\mathrm{orig}})$ after it has been rotated by an angle θ and translated by (δ_1,δ_2). Because the coordinates $\mathbf{x}'_{\mathrm{trans}}=T_\phi^{-1}(\mathbf{x}_{\mathrm{trans}})$ typically do not align with the original grid points, an interpolation algorithm is used to estimate the sound−speed values in the transformed image.

By parametrising the sound−speed image as a function of $\boldsymbol{\phi}$, we shift the optimisation process from updating individual pixels of $\mathbf{c}$ to adjusting the geometric transformation parameters. Consequently, the misfit function f depends on $\boldsymbol{\phi}$ solely through the transformed model

$$f(\boldsymbol{\phi})=f\left(\mathbf{c}_\phi\right),$$

enabling a more efficient search direction and a geometrically meaningful optimisation process.

Using the chain rule, the gradient of f with respect to the motion parameters is:

$$\frac{\partial f}{\partial \phi} = \left(\frac{\partial \mathbf{c}_\phi}{\partial \phi}\right)^T \frac{\partial f}{\partial \mathbf{c}_\phi}.$$

Here, $\partial f/\partial \mathbf{c}_\phi$ is the usual FWI gradient vector with respect to the sound speed (computed by the adjoint-state method), and $\partial \mathbf{c}_\phi/\partial \boldsymbol{\phi}$ is a Jacobian that describes how a change in the motion parameters $\boldsymbol{\phi}$ changes the sound speed $\mathbf{c}$ at each pixel.

This formula shows that we can obtain the gradient with respect to the motion parameters $\boldsymbol{\phi}$ by re-weighting and combining the values of the standard FWI gradient. If the manifold aspect of the optimisation process is neglected, the parameters may be updated directly using this gradient. Similarly, assuming small perturbations, the manifold can be considered locally flat and the Euclidean update is also recovered. In this work, we adopt a slightly more rigorous formulation by using the logarithmic and exponential maps from differential geometry. This strategy ensures that the angular component of SE(2) is bounded $-\pi \le \theta \le \pi$ and the translation component respects the curvature of the manifold. More detail, including the advantage relative to the flat-space approach, is provided in Appendix A.

Numerical implementation

The implementation is analogous to rigid registration. Each pixel in the sound−speed image is assigned a coordinate, which is a 2-D vector representing its position relative to a chosen centre of rotation. Given the current parameters $\boldsymbol{\phi} = \{\theta, \delta_1, \delta_2\}$, we construct the corresponding rigid transform T_ϕ :

1. The transformed sound−speed field $\mathbf{c}_\phi(\mathbf{x}'_{\text{trans}})$ is computed by mapping each grid point $\mathbf{x}_{\text{trans}}$ back to the original domain via $\mathbf{x}'_{\text{trans}} = T_\phi^{-1}(\mathbf{x}_{\text{trans}})$.
2. Bi-linear interpolation is used to evaluate $\mathbf{c}$ at coordinates $\mathbf{x}'_{\text{trans}}$ to form $\mathbf{c}_\phi$.
3. Solve the forward problem to obtain $L(\mathbf{c}_\phi)$.
4. Compute the data residual $L(\mathbf{c}_\phi) - \mathbf{d}$.
5. Solve the adjoint problem to form the standard FWI gradient vector $\partial f/\partial \mathbf{c}_\phi$.
6. Obtain the gradient with respect to the motion parameters by applying $\frac{\partial f}{\partial \phi} = \left(\frac{\partial \mathbf{c}_\phi}{\partial \phi}\right)^T \frac{\partial f}{\partial \mathbf{c}_\phi}$.

In practice, we do not provide an explicit expression for the Jacobian $\partial \mathbf{c}_\phi/\partial \boldsymbol{\phi}$. Instead, we use reverse-mode automatic differentiation (Pytorch [18]) to directly compute the product of this Jacobian transpose with the vector $\partial f/\partial \mathbf{c}_\phi$. The FWI gradient with respect to $\mathbf{c}$ is obtained using the adjoint-state method implemented in Stride [19].

The resulting gradient $\partial f/\partial \boldsymbol{\phi}$ gives the steepest descent direction in the space of motion parameters. We perform a gradient-based update of $\boldsymbol{\phi}$, with gradient normalisation and line search to improve stability near the minimum. This allows MOFI to capture the dominant geometric mismatch (translations and rotations) by using a more efficient and accurate search direction, in contrast with the pixel-wise update. Once this coarse alignment is achieved, the standard FWI gradient can, if desired, be used in the usual way to refine smaller-scale features.

The jobs were executed on a desktop computer with a 16-core CPU (2nd Gen Ryzen Threadripper 2 2950X, Advanced Micro Devices, Santa Clara, CA, USA), 128 GB RAM (Vengeance LPX DDR4 3200 MHz RAM, Corsair, Fremont, CA, USA) and 2 GPUs (Quadro8000 RTX 48GB, Nvidia, Santa Clara, CA, USA). Simulation runtimes were approximately 0.3 s per source *in silico* and 1.1 s per source *in vitro*. There were 64 sources per iteration *in silico* and 38 sources per iteration *in vitro*, with 3 simulations per source—required for the forward, adjoint and line search—with a total runtime of 4 h *in silico* and 8 h *in vitro*.

Results

In silico

Figure 1 (a) shows an *in silico* phantom representing the sound speed of the head. This phantom is constructed from the MIDA model [20] and the IT'IS database of tissue properties [21]. The shape of the grid is $(560, 450)$ and the isotropic spacing of the grid is 0.5 mm. Figure 1 provides an initial validation of our methodology. A large offset between (Fig. 1a) the correct and (Fig. 1b) the shifted p-wave velocity models causes (Fig. 1c) a large offset between the corresponding data (black and red traces). Figure 1 (d) shows that the standard FWI gradient is sensitive to this shift, but applying this gradient pixel-wise leads to inefficient updates. MOFI replaces pixel-wise updates with a small set of motion parameters (translations and rotations of the image), which provides a more efficient update direction. Figure 1 (e) is an initial validation of our hypothesis—the white arrows show the direction of the MOFI update, which is the reverse of the shift induced in Figure 1 (b).

Figure 2 demonstrates that MOFI is able to accurately locate the skull template when the ground truth sound speed undergoes a substantial shift in position and orientation (Fig. 2a). Compared with the original image, the rotation is 10° relative to the vertical about the centre of the image. The translation is $(14, -5.85)$mm.

Figure 2 (b) shows a starting model where the skull template has the correct sound speed and shape, but its position (*i.e.*, translation and rotation) is incorrect. Figure 2 (c) shows that FWI cannot reconstruct the brain from this starting model. Except for the position of the skull, this example is an inverse crime—a modelling situation where the reconstruction physics are identical to the simulation physics, the exact source wavelet and transducer positions are known and the data are not cycle-skipped as the inversion starts from 100 kHz .

Figure 2 (d) shows the three parameters $\boldsymbol{\phi}$ of SE(2) during the optimisation process. The translation starts at $(0, 0)$mm and the rotation starts at 0°. In total, there are 500 iterations. The last 200 iterations oscillate around a stationary point, which gives a translation $(13.9 \pm 0.3, -6.8 \pm 0.3)$mm and rotation of 9.8 ± 0.3°.

With respect to the known displacement, both components of translation are accurate to within 1 mm, and the rotation is accurate to within 1°. The stationary point is reached after approximately 200 iterations, which is computationally feasible. Furthermore, because the group has a limited number of parameters, the problem is—in a sense—overdetermined, which means that a small batch size (10% of the emitters) can be used. Figure 2 (e) shows the skull template at the optimal position, which is determined by the optimisation on SE(2). The FWI reconstruction in Figure 2 (f) confirms that the starting model is now correctly located.

In vitro

FWI reconstructions of laboratory data are substantially more challenging. First, the impulse response of the transducer is unknown. This means that the spatial response of the emitter and the sensitivity of the receiver are difficult to determine with precision. Second, the physics of the problem are unknown. To decrease computation time, we solve the acoustic wave equation with constant density and zero attenuation. In reality, materials have variable density and attenuate wave energy and may support elastic or anisotropic physics. Finally, the properties of the starting model are unknown. Different sources in the literature provide conflicting estimates of the skull sound speed, which suggests that these estimates have wide errors. Furthermore, the shape of the skull can only be determined imprecisely from CT or MRI data (typical resolution of 0.5 mm), and its position is only known approximately despite careful placement.

Figure 3 demonstrates that MOFI can optimise the position of the *in vitro* skull template. Figure 3 (column 1) shows the parameter values $\boldsymbol{\phi} = (\delta_1, \delta_2, \theta)$ for iterations 1, 50, 150, and 210. These correspond with the

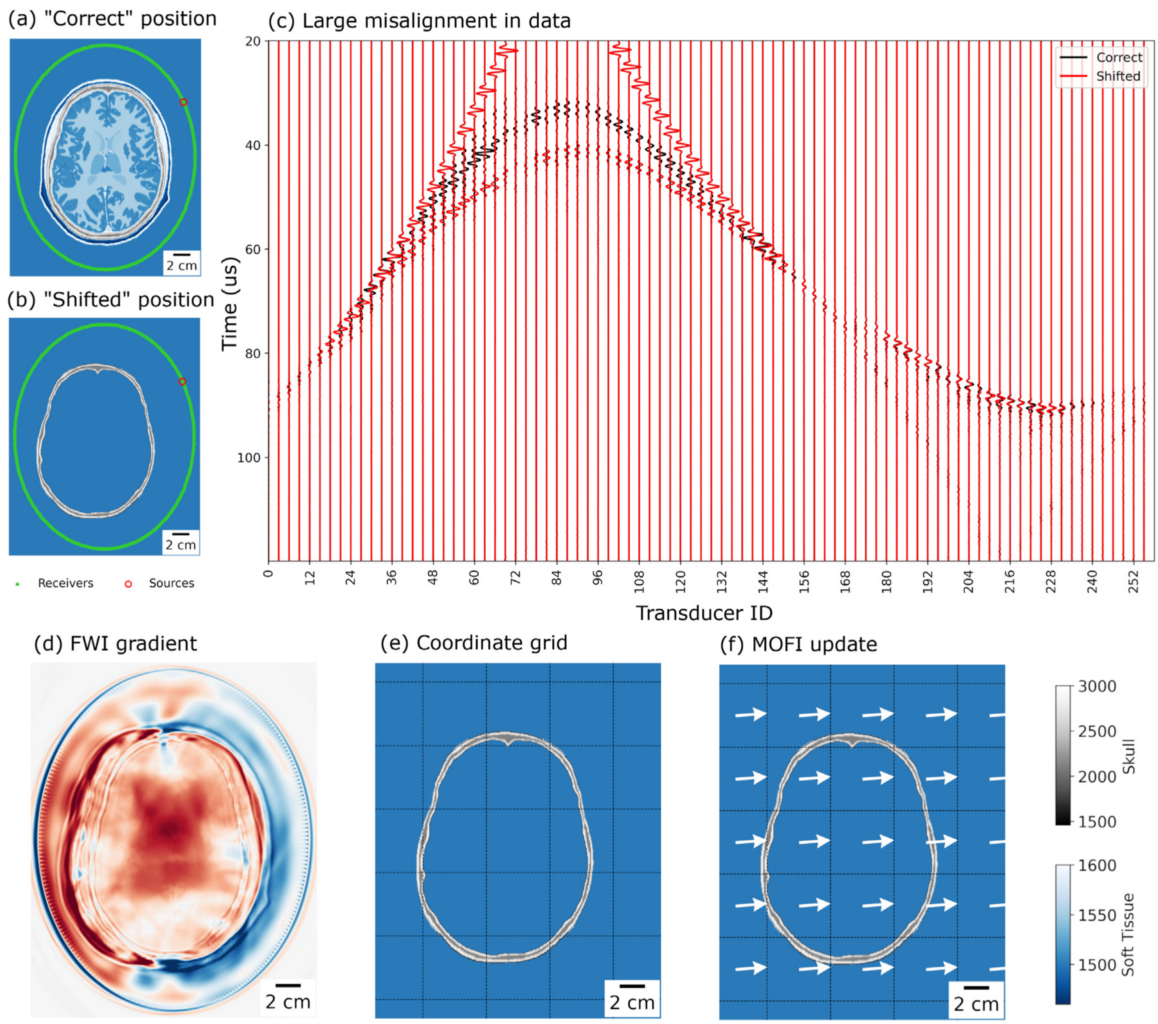

Figure 1. (a) The ground truth model in the "correct" position and (b) the skull template in the "shifted" position. (c) The correct and shifted data—respectively the *black* and *red traces*—show a large offset. (d) The full-waveform inversion (FWI) gradient captures the leftward shift, but the pixel-wise details are unsatisfactory. (d) By parametrising the image using coordinates, (f) manifold optimisation for FWI (MOFI) can represent the update in terms of translation and rotation of the image, providing an update direction (*white arrows*) that captures the shift.

sound speed images in Figure 3 (column 2), which show the template moving smoothly from the initial to the final position.

Figures 3 (columns 3−5) provide a detailed view of the data misfit during the optimisation process. The observed acoustic data are in black, and the predicted acoustic data are in red.

Broadly, transmitted energy in the predicted data (selection 2, red traces) is smoothly shifted to adjacent receivers, but the arrival time remains unchanged. In contrast, reflected energy (selections 1 and 3, red traces, second arrival) is shifted to earlier arrival times. The first arrival in selections 1 and 3 travels directly through water and remains unchanged throughout the optimisation. In detail:

- Figure 3, iteration 1: The predicted and observed data exhibit significant misalignment across all selections.
- Figure 3, iteration 50: In selections 1 and 3, the reflected predicted data (second arrivals) shift to earlier times, moving toward the observed data. In selection 3, a complete alignment in phase and amplitude is achieved, while the transmitted data in selection 2 remain largely unchanged.
- Figure 3, iteration 150: In selection 3, the predicted reflected data overshoot the observed data, whereas in selection 1 (on the opposite side of the array) the arrival time continues to steadily decrease. In selection 2, the timing of the transmitted data remains stable, although the energy is redistributed to adjacent receivers.
- Figure 3, iteration 210: The data in all three selections align closely: transmitted energy is correctly redistributed, and reflected energy is appropriately time-shifted. Minor misalignments persist, particularly for reflected arrivals in selection 1, which are cycle-skipped.

Figure 4 demonstrates that MOFI identifies an accurate position and orientation for the skull template. Figure 4 (a) shows that the optimal translation is $(7.2 \pm 0.2, 0.7 \pm 0.2)$mm and the optimal rotation is $10.71 \pm 0.07^{\circ}$, measured from the final 300 iterations. Figure 4 (b) shows the loss or data fit during the inversion, which is noisy as only 10% of the 384 sources were used to calculate the gradient. The optimisation process converges after iteration 200, evidenced by stable translation and rotation values and a flattened loss curve. Using the full dataset (*i.e.*, all sources) the loss per source at the starting position is is 398.9 and 375.9 at the final position, a percentage reduction of 5.7%.

Figure 4 (c) shows the skull template in its initial position. Previous work [8] shows that this template has approximately the correct sound speed and shape, but none of the values are known precisely. Figure 4 (d) confirms the incorrect placement because the FWI reconstruction has failed. Figure 4 (e) shows the skull template in the optimal position,

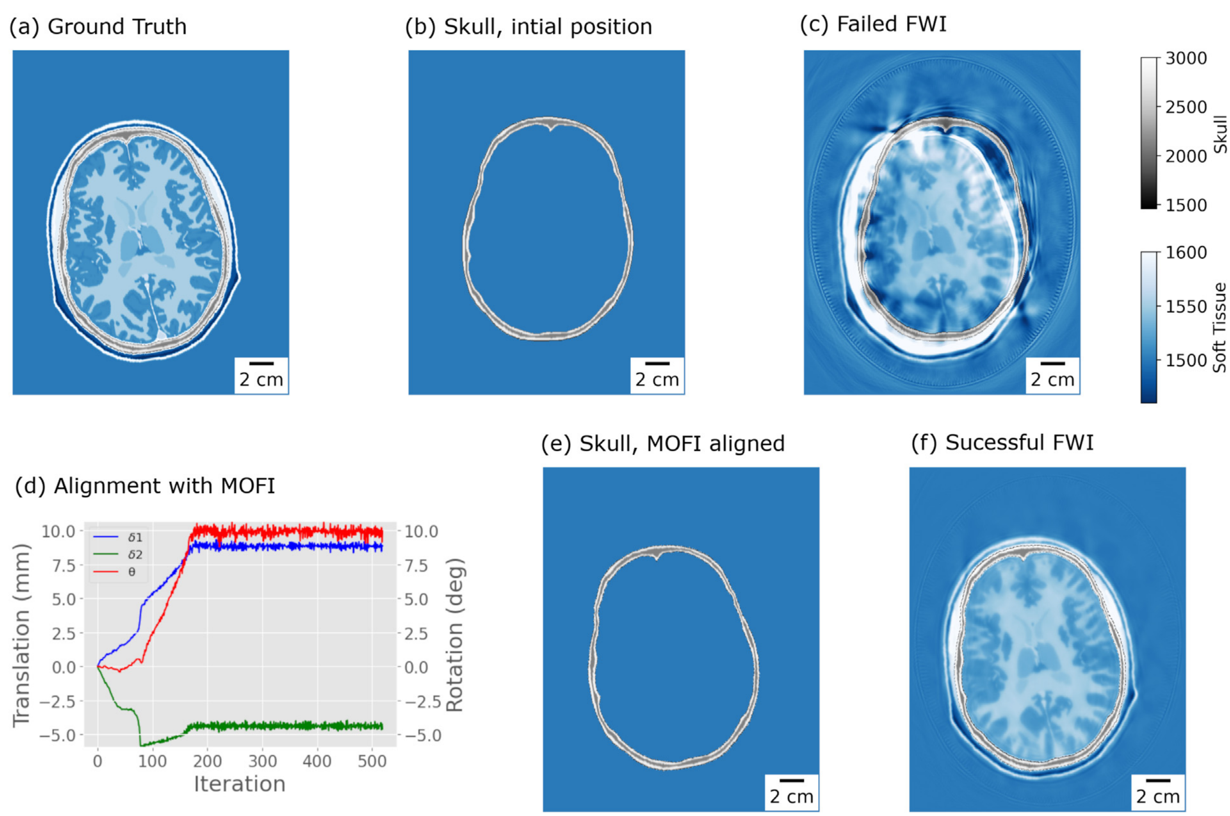

Figure 2. *In silico* wavefield data are simulated from (a) the ground truth sound speed. (b) The skull template at the initial position produces (c) a failed full-waveform inversion (FWI) reconstruction. (d) Manifold optimisation for FWI (MOFI) identifies the correct translation ($\delta 1$, $\delta 2$) and rotation (θ). (d) The MOFI aligned skull template produces (f) a successful FWI reconstruction.

identified at iteration 500. Figure 4 (f) shows an accurate FWI reconstruction, which confirms that the starting model is now correctly positioned.

Discussion

Our approach delivers strong results whenever an approximate sound−speed template is available and requires alignment. Ultrasound neurotechnologies rely on accurate skull template alignment; misalignment compromises the precision of numerical refocusing corrections. Rigid MOFI addresses this challenge by enabling automatic skull template alignment with subpixel accuracy. The *in vitro* results presented in Figure 4 demonstrate that this level of accuracy is likely to be sufficient for FWI neuroimaging. Similarly, a sensitivity analysis by Robertson *et al*. [6] suggests that errors of approximately 1 mm are tolerable for focused ultrasound examinations, a threshold our results indicate is achievable with MOFI. This innovation reduces the need for manual intervention in FWI neuroimaging and provides a robust alternative—or complement—to MRI-guided image registration in transcranial focused ultrasound therapy.

The effectiveness of MOFI stems from its use of the Special Euclidean Group SE(2) to constrain the FWI optimisation process. SE(2) is part of a broader class of mathematical structures known as Lie Groups, which combine algebraic group properties with the smooth geometry of manifolds. These groups are particularly suited to modelling continuous transformations—such as rotations and translations—while preserving geometric symmetries like angles and distances. As a result, Lie groups offer a natural framework for optimisation problems. Their utility is well documented in fields such as image registration and robotics. For instance, diffeomorphic non-rigid image registration, which leverages the diffeomorphism group, significantly outperforms freeform deformation methods when handling large deformations [22]. In robotics, manifold optimisation using Lie groups enhances the accuracy of trajectory mapping [23−25].

Another perspective emerges from convex optimisation, where computational complexity scales polynomially with the number of dimensions [26]. Although FWI is non-convex, the convex case implies that the complexity of non-convex optimisation also increases with dimensionality. Conventional FWI optimisation algorithms operate on a discretised sound−speed grid, where the grid is designed to ensure the stability of the numerical simulator rather than optimisation efficiency. In Figures 2 and 4, the discretised sound-speed grids consist of 252,000 and 672,000 pixels, respectively—each pixel representing an individual dimension in the optimisation space.

Thus, by constraining the optimisation to just three dimensions—rotation and translation—MOFI circumvents the challenges posed by high-dimensional discretised grids, such as those in Figures 2 and 4, where each pixel introduces an additional optimisation variable. Dimensionality reduction by manifold optimisation is not unique to FWI [14,27,28] but, through the use of SE(2), the Lie group structure of MOFI ensures a smooth optimisation manifold that retains algebraic group properties. This finding contrasts with level set methods [27], which lack a group structure and may struggle to enforce specific deformations. Furthermore, unlike methods that invert for finite−element grid control points [14,28], MOFI guarantees invertibility, preventing grid points from self-intersecting and preserving the image's topology. As a result, MOFI delivers a robust and efficient optimisation process, readily adaptable to any discretisation (*e.g.*, finite−difference or finite −element) while maintaining stability for large deformations.

Looking ahead, we aim to explore more expressive Lie groups, such as the diffeomorphism group, and investigate how concepts from Lie groups could enhance deep learning approaches. The design of deep learning models often results in non-smooth representations—exemplified by the challenges in training generative adversarial networks.

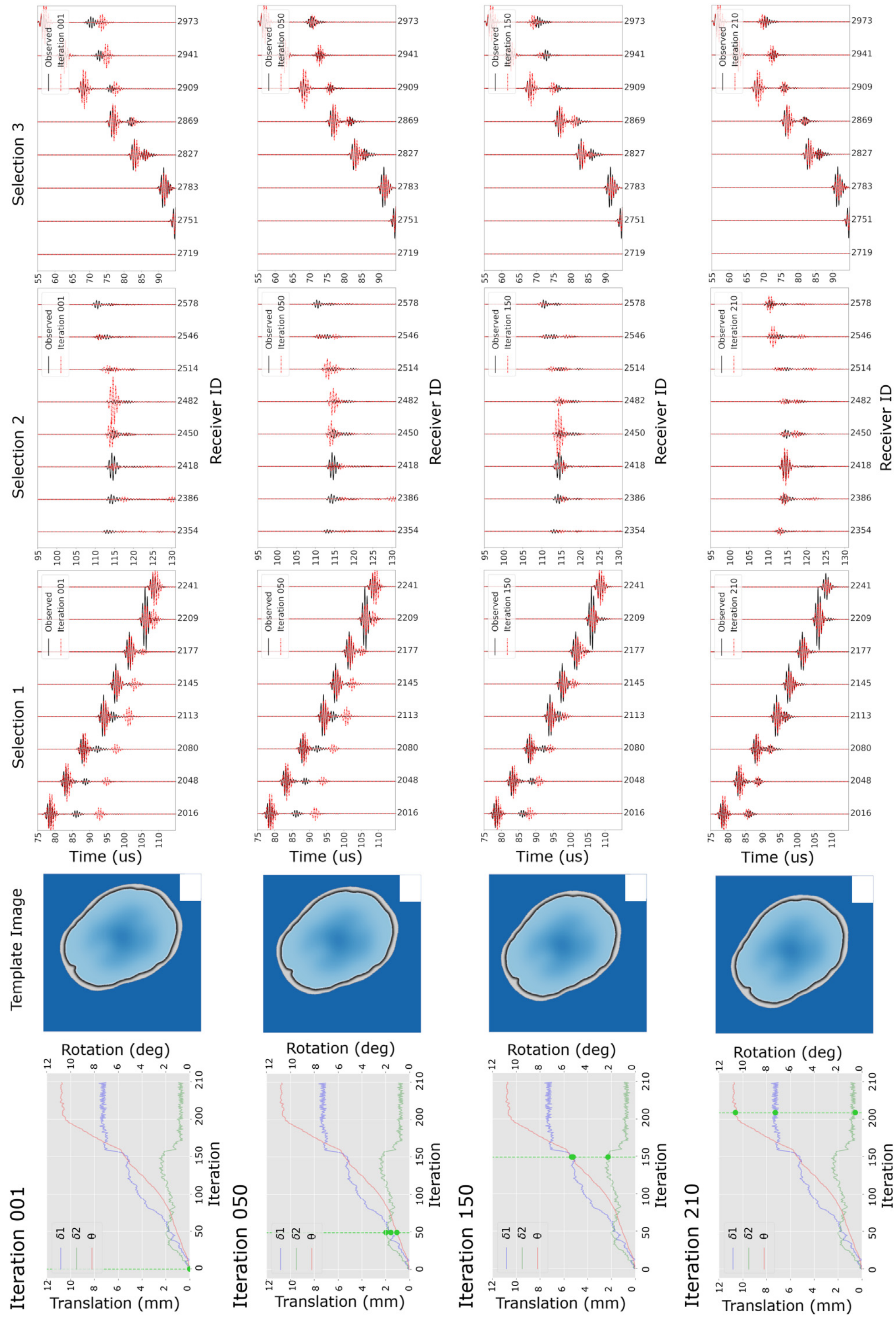

Figure 3. Manifold optimisation for FWI (MOFI) optimisation on the *in vitro* skull template. (*Column 1*) The parameters ($\delta 1$, $\delta 2$, θ) follow a smooth curve from the initial to the final values, which (*column 2*) corresponds with a smooth movement from the initial to the final position. (*Column 3, selection 1*) One set of the reflected data are gradually time-shifted to earlier. (*Column 4, selection 2*) The transmitted data are redistributed to adjacent transducers, but not time shifted. (*Column 5, selection 3*) The second set of reflected data are also time shifted, but these data overshoot before setting at the correct delay.

Approaches that regularise inverse problems but do not explicitly enforce smoothness, such as the deep image prior [29] or score-based generative [30] models, may therefore benefit from incorporating this principle. Similarly, normalising flows, which use smooth, invertible transformations, have already demonstrated their utility in FWI [13,31].

It is well documented that FWI struggles to converge when the predicted and observed data arrive out of phase [10]. MOFI is much more resilient than the typical requirement that the phase difference be less than 90°, but it will eventually struggle if this distance becomes too large. Therefore, MOFI also benefits from the multi-scale method [32]. Similarly, the current loss function uses the L2 norm, but alternative formulations from the FWI literature could further expand the basin of attraction.

In terms of optimisation parameters, both fixed-step and explicit line-search methods were evaluated, with explicit line-search providing greater stability at the optimum. Gradient normalisation was found to

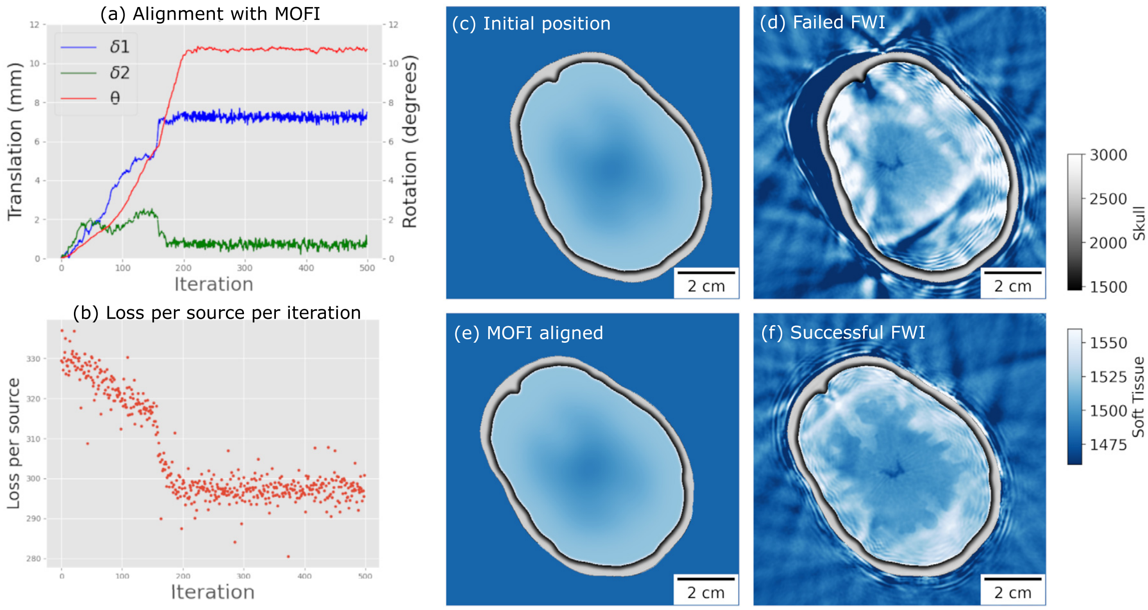

Figure 4. (a) Manifold optimisation for full-waveform inversion (MOFI) identifies a translation $\boldsymbol{\delta} = (7.2 \pm 0.2, 0.7 \pm 0.2)$mm and a rotation $\theta = 10.71 \pm 0.07^{\circ}$. (b) The loss used to calculate the gradient and normalised per source. (c) The skull template at the initial position produces (d) a failed full-waveform inversion (FWI) reconstruction. (e) The MOFI-aligned skull template provides (f) a successful FWI reconstruction.

be essential, and using the norm of the first gradient proved most effective, whereas normalising by the maximum of the current gradient led to instability.

The physics used assumes homogeneous density with an absence of viscous or elastic effects. Exclusion of these physics is known to affect the quality of FWI reconstruction and their degree of influence by MOFI has not been studied. The *in vitro* dataset was designed to replicate an in vivo neuroimaging system [8]; materials with different densities were used and elastic effects are clearly visible in the observed dataset, which provides some confidence that MOFI can tolerate these effects. A future study to extend MOFI to 3-D in vivo datasets will establish their influence with certainty.

Conclusions

Our findings demonstrate that a skull sound−speed template can be accurately and automatically located using only acoustic data. This strategy eliminates the need for manual alignment of the starting model in ultrasound neuroimaging and potentially enhances the image-to-image alignment used during ultrasound therapy. We greatly improve the convergence properties of FWI by leveraging the special Euclidean Lie group, which reduces the dimensionality of the FWI optimisation problem while preserving a smooth optimisation manifold. This strategy has been successful in other fields, particularly nonrigid image registration, and suggests that more expressive Lie groups could entirely eliminate the need for patient-specific prior information in wave-based image reconstruction algorithms. Together, these findings support the use of MOFI as a principled pathway toward practical, guidance-free transcranial ultrasound tomography.

Acknowledgments

The authors thank the anonymous reviewers for their time and effort in reviewing this work. The authors thank the team at Sonalis Imaging Limited for their support and insightful discussions.

This work was funded by EPSRC grant number EP/X033651/1. Oscar Calderon-Agudo was funded by MRC, Future Leader Fellowship (FLF), MR/V024086/1.

During the preparation of this work the author(s) used Le Chat Mistral to assist with drafting the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Appendix A. Implementing manifold updates

In flat Euclidean space, updates to the transformation parameters $\boldsymbol{\phi}$ are applied directly as additive increments,

$$\boldsymbol{\phi}^{k+1} = \begin{pmatrix} \theta^k + \Delta_\Theta \\ \delta_1^k + \Delta_{\delta_1}, \\ \delta_2^k + \Delta_{\delta_2} \end{pmatrix},$$

where index k denotes the current iteration, $k+1$ the updated iteration and Δ_ϕ the update with direction $-\partial f/\partial\phi$.

In contrast, a manifold update respects the curved geometry of SE(2). Among the possible formulations of curved-space updates (*e.g.*, Example 7, [25]), we adopt a representation based on the product manifold $\mathbb{R}^2 \times$ SO(2). Here, rotations and translations are treated as independent manifolds, but their interaction is preserved through the group structure. This formulation is computationally convenient and provides a natural comparison with the flatspace update.

The manifold update leverages the logarithmic and exponential maps, which extend the familiar logarithmic and exponential functions to matrices [33]. These maps facilitate projections between the group element $g \in \mathrm{SE}(2)$ and its Lie algebra $\tau \in \mathscr{S}\mathscr{E}(2)$. For g defined as

$$g = \begin{pmatrix} \cos(\theta^k) & -\sin(\theta^k) & \delta_1^k \\ \sin(\theta^k) & \cos(\theta^k) & \delta_2^k \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} R^k & \delta^k \\ 0 & 1 \end{pmatrix},$$

the update process begins with the pullback of the group element into the Lie algebra using the logarithmic map. For the product manifold, the update separates into translation and rotation components.

The rotation update uses the logarithmic map from the special orthogonal group, projecting the update from the manifold $R \in \mathrm{SO}(2)$ to the Lie algebra $\tau_\Theta \in \mathscr{S}\mathscr{V}(2)$,

$$\tau_\Theta = \log(\Delta_R) = \mathrm{atan2}(\Delta_{R_{21}}, \Delta_{R_{11}}),$$

where $\Delta_{R_{21}}$ and $\Delta_{R_{11}}$ are the off-diagonal and diagonal elements of the rotation update matrix Δ_R, respectively. This projection ensures the rotation update is represented in the algebra, where it can be applied linearly,

$$\theta^{k+1} = \theta^k + \tau_\theta.$$

Note that this simple update is only possible because the elements of SO(2) commute. If the group elements were non-commutative, then the Lie algebra update would need to use the Baker−Campbell−Hausdorff series [34, page 125].

Finally, the pushforward of the updated rotation moves back to the manifold using the exponential map,

$$R^{k+1} = \mathrm{Exp}\left(\theta^{k+1}\right) = \begin{pmatrix} \cos\left(\theta^{k+1}\right) & -\sin\left(\theta^{k+1}\right) \\ \sin\left(\theta^{k+1}\right) & \cos\left(\theta^{k+1}\right) \end{pmatrix}.$$

The translation update does not use the logarithmic or exponential map, because the translation manifold is Euclidean and requires no projection. However, composition of two elements in $\mathbb{R}^2 \times \mathrm{SO}(2)$ does adjust the direction of the translation update,

$$\delta^{k+1} = \delta^k + R^k \Delta_\delta.$$

Again, for small perturbations where $R^k \approx I$, this reduces to the flat space update.

Another distinction between the manifold and flatspace updates lies in the range of the rotation angle. In the manifold formulation, θ is constrained to the canonical range $-\pi \leq \theta \leq \pi$. The importance of this distinction is evident when interpolating between angles. For example, traversing from 10° to −10° in flat space would follow the long path $10^\circ \leq \theta \leq 350^\circ$ (a 340° span), whereas the manifold formulation takes the shorter, geometrically meaningful path $10^\circ \geq \theta \geq -10^\circ$ (a 20° span).

References

[1] Estrada H, Deffieux T, Robin J, Tanter M, Razansky D. Imaging the brain by traversing the skull with light and sound. Nat Biomed Engineer 2025;9:1574–90.
[2] Tufail Y, Matyushov A, Baldwin N, Tauchmann ML, Georges J, et al. Transcranial pulsed ultrasound stimulates intact brain circuits. Neuron 2010;66:681–94.
[3] Hynynen K, McDannold N, Vykhodtseva N, Jolesz FA. Noninvasive MR imaging-guided focal opening of the blood-brain barrier in rabbits. Radiology 2001;220:640–6.
[4] Elias WJ, Huss D, Voss T, Loomba J, Khaled M, Zadicario E, et al. A pilot study of focused ultrasound thalamotomy for essential tremor. N Engl J Med 2013;369:640–8 _eprint: https://www.nejm.org/doi/pdf/10.1056/NEJMoa1300962.
[5] Pichardo S, Sin VW, Hynynen K. Multifrequency characterization of the speed of sound and attenuation coefficient for longitudinal transmission of freshly excised human skulls. Phys Med Biol 2010;56:219.
[6] Robertson J, Martin E, Cox B, Treeby BE. Sensitivity of simulated transcranial ultrasound fields to acoustic medium property maps. Phys Med Biol 2017;62:2559.
[7] Guasch L, Calderon Agudo O, Tang M-X, Nachev P, Warner M. Full-waveform inversion imaging of the human brain. NPJ Digit Med 2020;3:1–12.
[8] Robins TC, Cueto C, Cudeiro J, Bates O, Agudo OC, Strong G, et al. Dual -probe transcranial full-waveform inversion: a brain phantom feasibility study. Ultrasound Med Biol 2023;49:2302–15.
[9] Mitcham T, Ali R, Singh M, McConnell S, Rahmani R, Schartz D, et al. Transcranial ultrasound tomography for brain imaging: ex vivo results and potential for stroke imaging. Med Phys 2025;52:e18090.
[10] Virieux J, Operto S. An overview of fullwaveform inversion in exploration geophysics. Geophysics 2009;74:WCC1–26.
[11] van Leeuwen T, Herrmann FJ. Mitigating local minima in full-waveform inversion by expanding the search space. Geophys J Int 2013;195:661–7.
[12] Warner M, Guasch L. Adaptive waveform inversion: theory. Geophysics 2016;81:R429–45.
[13] Orozco R, Witte P, Louboutin M, Siahkoohi A, Rizzuti G, Peters B, et al. Invertible-Networks.jl: a Julia package for scalable normalizing flows. J Open Source Softw 2024;9:6554. ADS Bibcode: 2024JOSS....9.6554O.
[14] Marty P, Boehm C, Fichtner A. Shape optimization for transcranial ultrasound computed tomography. Medical Imaging 2023: Ultrasonic Imaging and Tomography. Bellingham, WA: SPIE; 2023. p. 77–88.
[15] Aubry J-F, Tanter M, Pernot M, Thomas J-L, Fink M. Experimental demonstration of noninvasive transskull adaptive focusing based on prior computed tomography scans. J Acoust Soc Am 2003;113:84–93.
[16] McDannold N, Clement GT, Black P, Jolesz F, Hynynen K. Transcranial magnetic resonance imaging-guided focused ultrasound surgery of brain tumors: initial findings in 3 patients. Neurosurgery 2010;66:323.
[17] O'Reilly MA, Jones RM, Birman G, Hynynen K. Registration of human skull computed tomography data to an ultrasound treatment space using a sparse high frequency ultrasound hemispherical array. Med Phys 2016;;43(9)):5063–71 eprint https://aapm.onlinelibrary.wiley.com/doi/pdf/10.1118/1.4960362.
[18] Paszke A, Gross S, Massa F, Lerer A, Bradbury J, Chanan G, et al. PyTorch: an imperative style, high-performance deep learning library. Advances in Neural Information Processing Systems 32. Red Hook, NY: Curran Associates; 2019.
[19] Cueto C, Bates O, Strong G, Cudeiro J, Luporini F, Agudo Calderón, Gorman G, et al. Stride: a flexible software platform for high-performance ultrasound computed tomography. Comput Methods Programs Biomed 2022;221:106855.
[20] Iacono MI, Neufeld E, Akinnagbe E, Bower K, Wolf J, Oikonomidis IV, et al. MIDA: a multimodal imagingbased detailed anatomical model of the human head and neck. PLoS One 2015;10:e0124126.
[21] P Hasgall, E Neufeld, M Gosselin, A Klingenbock, N Kuster. IT'IS Database for thermal and electromagnetic parameters of biological tissues. https://api.semanticscholar.org/CorpusID:139791227; 2015.
[22] Beg MF, Miller MI, Trouvé A, Younes L. Computing large deformation metric mappings via geodesic flows of diffeomorphisms. Int J Computer Vis 2005;61:139–57.
[23] Brockett RW. Robotic manipulators and the product of exponentials formula. In: Fuhrmann PA, editor. Mathematical theory of networks and systems. Berlin, Heidelberg: Springer; 1984.. p. 120–9.
[24] Horn BKP. Closed-form solution of absolute orientation using unit quaternions. JOSA A 1987;4:629–42.
[25] J Solà, J Deray, D Atchuthan. A micro Lie theory for state estimation in robotics. 2021. arXiv:1812.01537 [cs].
[26] Boyd SP, Vandenberghe L. Convex optimization. Cambridge, UK: Cambridge University Press; 2004 Google-Books-ID: mYm0bLd3fcoC.
[27] Kadu A, van Leeuwen T, Mulder WA. Salt reconstruction in full-waveform inversion with a parametric level-set method. IEEE Trans Computat Imaging 2017;3:305–15.
[28] Shi F, Huthwaite P. Waveform -based geometrical inversion of obstacles. Phys Rev Appl 2019;12:064008.
[29] Ulyanov D, Vedaldi A, Lempitsky V. Deep image prior. arXiv:1711.10925 [cs.CV].
[30] Y Song, L Shen, L Xing, S Ermon. Solving inverse problems in medical imaging with score-based generative models. 2022. arXiv:2111.08005 [eess].
[31] Sun C, Malcolm A, Kumar R, Mao W. Enabling uncertainty quantification in a standard full-waveform inversion method using normalizing flows. Geophysics 2024;89:R493–507.
[32] Bunks C, Saleck FM, Zaleski S, Chavent G. Multiscale seismic waveform inversion. Geophysics 1995;60:1457–73.
[33] Moler C, Van Loan C. Nineteen dubious ways to compute the exponential of a matrix, twenty-five years later. SIAM Review 2003;45(1)):3–49 _eprint:. doi: 10.1137/S00361445024180.
[34] Tapp K. Matrix groups for undergraduates, vol. 29 of The Student Mathematical Library. Providence, RI: American Mathematical Society; 2005.